

# AIMS System - Detailed Class Design Documentation

## 1. Design Class: SearchController

Table 1. Attribute design

| # | Name              | Data type     | Default value | Description                                   |
|---|-------------------|---------------|---------------|-----------------------------------------------|
| 1 | productList       | List<Product> | empty list    | List of products in the system                |
| 2 | searchResults     | List<Product> | Empty list    | Products matching search                      |
| 3 | maxResultsPerPage | int           | 20            | Maximum number of products displayed per page |

Table 2. Operation design

| # | Name          | Return type   | Description (purpose)                |
|---|---------------|---------------|--------------------------------------|
| 1 | searchKeyWord | List<Product> | Search for products based on keyword |
| 2 | searchProduct | List<Product> | Perform search within product list   |

Parameter:

- **searchKeyWord(keyword: String): List<Product>**
  - **keyword:** String keyword input from HomeScreen
  - **Return:** List of products matching the keyword
- **searchProduct(keyword: String): List<Product>**
  - **keyword:** String keyword to compare with product attributes
  - **Return:** List of products matching the keyword

Exception:

- **searchKeyWord(keyword: String): List<Product>**
  - **EmptyKeywordException:** Thrown if search keyword is empty or null

Method:

- **searchKeyWord(keyword: String): List<Product>**
  - **Purpose:** This method receives keywords from HomeScreen and processes the search
  - **Processing:**
    1. Validate keyword
    2. Call searchProduct method with provided keyword
    3. Return search results limited by maxResultsPerPage
  - **Return:** List of products matching the keyword, limited to 20 products per page
- **searchProduct(keyword: String): List<Product>**
  - **Purpose:** Perform product search based on keyword
  - **Processing:**
    1. Iterate through the product list
    2. Compare keyword with product attributes (name, description, type, etc.)
    3. Collect matching products
  - **Return:** List of all products matching the keyword

State:

- When no search has been performed, searchResults is an empty list
- After search, searchResults contains products matching the keyword
- If no matching products are found, searchResults is an empty list